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SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme: **B.E / B.Tech**
Course Code: **MLA02**
Course Name: **FUNDAMENTALS OF MACHINE LEARNING**
Duration: **1:30 Hr**
Date : _____ Max Marks: **50 Marks**

ASSIGNMENT – I

UNIT I – INTRODUCTION AND MATHEMATICAL FOUNDATIONS

Topics: *Design of a learning system, types of machine learning, probability theory and distributions, Bayes decision theory, information theory*

Course Outcome

On successful completion of this assignment, students will be able to:

CO	Course Outcome Statement
CO1	Analyse machine learning fundamentals and mathematical foundations, including probability theory, Bayes decision theory, and information theory.

Answer all the Tasks

Task 1. A retail bank plans five machine-learning systems. For each system, analyse the problem and state (i) the learning paradigm (supervised / unsupervised / semi-supervised / reinforcement), (ii) the learning task (classification, regression, clustering, control, etc.) and (iii) the experience E and performance measure P in Mitchell's (T, P, E) formulation:

(a) Predict whether a loan applicant will default, using ten years of labelled repayment records.

(b) Discover behavioural customer segments among 2 million customers for which no labels exist.

(c) Estimate the next-month credit-card spend (in ₹) of each customer.

(d) Learn an ATM cash-refill policy that maximises cash availability minus transport cost through repeated trial and feedback.

(e) Only 2% of 500,000 transaction records are labelled fraud/genuine; the rest are unlabelled. Build a fraud detector that exploits both kinds of data.

Finally, justify which of the five systems is most harmed by label noise.

(5) (CO1) (BL4)

Task 2. A screening test for a disease with prevalence 2% has sensitivity 95% and specificity 90%. Calculate the probability that a person who tests positive actually has the disease, and analyse why the answer is much lower than the sensitivity.

(2) (CO1) (BL4)

Task 3. The joint probability mass function of a discrete random vector (X, Y) is given below.

X \ Y	Y = 0	Y = 1
X = 0	0.10	0.20
X = 1	0.15	0.25

Course Code: MLA0202
Course: Fundamentals of Machine Learning

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1) Task 1: Design of Learning Systems

System	Learning paradigm	Learning Task	Experience (E)	Performance (P)
(a) Loan default	Supervised	Classification	10 years of labelled repayment records	Accuracy, F1-score
(b) Customer Segments	Unsupervised	Clustering	2 million customer records	Silhouette score
(c) Credit card spend	Supervised	Regression	Historical spending data	MSE, RMSE
(d) ATM Cash refill	Reinforcement	Control/ Decision making	Repeated trials and feedback	Total reward
(e) Fraud detection	Semi-Supervised	Classification	2% labelled and 99% unlabelled records	Precision, Recall, F1-score

Label noise: System (a) is directly affected by incorrect repayment labels

2) Bayes' Theorem

$$P(+|D) = 1 - 0.90 = 0.10$$

$$P(D|+) = \frac{P(+|D)\,P(D)}{P(+|D)\,P(D) + P(+|\bar{D})\,P(\bar{D})}$$

$$= \frac{0.95(0.02)}{0.95(0.02) + 0.10(0.98)}$$

$$= \frac{0.019}{0.117} = 0.1624$$

$$\Rightarrow P(D|+) = 16.24\%$$

3) Joint Probability Distribution

(a) For X: $P(X=0) = 0.10 + 0.20 = 0.30$

$P(X=1) = 0.15 + 0.25 = 0.40$

$P(X=2) = 0.20 + 0.10 = 0.30$

For Y: $P(Y=0) = 0.10 + 0.15 + 0.20 = 0.45$

$P(Y=1) = 0.20 + 0.25 + 0.10 = 0.55$

(b) $E[X] = 0(0.3) + 1(0.4) + 2(0.3) = 1$

$E[X^2] = 0 + 1(0.4) + 4(0.3) = 1.6$

$Var(X) = E[X^2] - [E[X]]^2 = 1.6 - 1$

$Var(X) = 0.6$

$E[Y] = 0(0.45) + 1(0.55) = 0.55$

$$P(X=2|Y=1) = \frac{P(X=2, Y=1)}{P(Y=1)} = \frac{0.10}{0.55} = 0.1818$$

(d) $E[XY] = 0 + 0.25 + 0.20 = 0.45$

$Cov(X,Y) = E[XY] - E[X]E[Y] = 0.45 - (1)(0.55) = -0.10$

$Var(Y) = 0.55(0.45) = 0.2475$

$$r = \frac{Cov(X,Y)}{\sqrt{Var(X)Var(Y)}} = \frac{-0.10}{\sqrt{0.6(0.2475)}} = -0.2594$$

(e) $P(X=0, Y=0) = 0.10$

$P(X=0)P(Y=0) = 0.30(0.45) = 0.135$

$\therefore \quad 0.10 \neq 0.135$

X and Y are not independent

4) Normal Distribution and Maximum Likelihood Estimation

(a) $$Z = \frac{58-50}{4} = 2$$

$P(X > 58) = 1 - \Phi(2) = 1 - 0.9772 = 0.0228$

$$Z_1 = \frac{46-50}{4} = -1$$

$$Z_2 = \frac{54-50}{4} = 1$$

$P(-1 < Z < 1) = 0.8413 - 0.1587 \Rightarrow P(46 < X < 54) = 0.6826$

(b) $$\hat{\mu} = \frac{12+15+11+14+13}{5} = 13$$

$$\hat{\sigma}^2_{ML} = \frac{\sum (x_i - \hat{\mu})^2}{n} = \frac{1+4+4+1+0}{5} = 2$$

$$s^2 = \frac{10}{5-1} = 2.5$$

ML variance is biased because it divides by n instead of n-1. The unbiased variance is 2.5

5) Information Theory - Entropy

$$H(X) = -\sum p_i \log_2 p_i = -\left[\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{4}\log_2\frac{1}{4} + 2\left(\frac{1}{8}\log_2\frac{1}{8}\right)\right]$$

$$= 0.5 + 0.5 + 0.375 + 0.375$$

$$= 1.75 \text{ bits}$$

$$H(X) = \log_2 4 = 2 \text{ bits}$$

$$R = 2 - 1.75 = 0.25 \text{ bits}$$

Percentage redundancy, $\frac{0.25}{2} \times 100 = 12.5\%$

$$L = \sum p_i l_i = \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{8}(3)$$

$$= 1.75 \text{ bits}$$

6) KL Divergence

$$D_{KL}(P||Q) = \sum P_i \log_2 \frac{P_i}{Q_i} = 0.5\log_2(0.5) + 0.25\log_2(0.75) + 0.25\log_2(0.75)$$

$$= 0.0850 \text{ bits}$$

$$D_{KL}(Q||P) = \sum Q_i \log_2 \frac{Q_i}{P_i} = \frac{1}{3}\log_2\frac{2}{3} + \frac{1}{3}\log_2\frac{4}{3} + \frac{1}{3}\log_2\frac{4}{3}$$

$$= 0.0817 \text{ bits}$$

∴ KL Divergence is asymmetric

$$D_{KL}(P||Q) \neq D_{KL}(Q||P)$$

7) Entropy and Mutual Information

$P(A=0) = 0.5$, $P(A=1) = 0.5$

$H(A) = -0.5\log_2 0.5 - 0.5\log_2 0.5$

$H(A) = 1$ bit

$\Rightarrow H(B) = 1$ bit

$$H(A,B) = -\sum p(a,b)\log_2 p(a,b) = -2(0.4\log 0.4) - 2(0.1\log_2 0.1)$$
$$= 1.7219 \text{ bits}$$

$$H(A|B) = H(A,B) - H(B) = 1.7219 - 1 = 0.7219 \text{ bits}$$

$$I(A;B) = H(A) - H(A|B) = 1 - 0.7219 = 0.2781 \text{ bits}$$

B provides some information about A and can be used as a predictive feature, but it does not completely determine A.

8) Bayes Decision Theory

(a) $$x^* = \frac{\mu_1+\mu_2}{2} + \frac{\sigma^2}{\mu_2-\mu_1}\ln\frac{P(\omega_1)}{P(\omega_2)} = 4 + \frac{4}{4}\ln\frac{0.6}{0.4} = 4 + \ln(1.5)$$
$$= 4.4055$$

(b) $4.5 > 4.4055$ [Classify as ω_2]

(c) $$P(\text{error}) = P(\omega_1)P(x > x^*|\omega_1) + P(\omega_2)P(x < x^*|\omega_2)$$
$$= 0.6[1 - \phi(1.2027) + 0.4\,\phi(-0.7973)$$
$$\approx 0.6(0.1146) + 0.4(0.2127)$$
$$\approx 0.154$$

$\Rightarrow P(\text{error}) = 15.4\%$

Threshold movement: As the prior probability of ω_1 increases, the threshold moves towards ω_2.

9) Minimum Risk Classification

$$R(\alpha_1|x) = 0(0.3) + 8(0.7) = 5.6$$
$$R(\alpha_2|x) = 50(0.3) + 0(0.7) = 15$$

$5.6 < 15$

Choose α_1 (Declare disease)

$0.7 > 0.3$

MAP chooses α_2

$$8(1-p) < 50p$$

$$8 < 58p$$

$$p > \frac{8}{58} = 0.1379$$

10) Bias – Variance Trade-off

Gap = Validation MSE - Training MSE

Degree	Calculation	Gap
1	9.5 - 9.0	0.5
2	4.6 - 1.1	0.5
3	2.9 - 2.2	0.7
5	3.6 - 0.9	2.7
9	11.8 - 0.1	11.7

Degree 1 : Underfitting

Degree 2 : Better fit

Degree 3 : Best generalization

Degree 4 : Overfitting begins

Degree 5 : Severe overfitting

Best Model : Degree 3

Validation MSE is minimum at 2.9

Improvement:

- Apply regularisation (L1/L2).
- Use more training data or reduce polynomial complexity.

Bayesian Estimation

(a)

$$\hat{\theta}_{ML} = 7/10 = 0.70$$

$$\hat{\theta}_{MAP} = \frac{\alpha + H - 1}{a+b+n-2} = \frac{2+7-1}{2+2+10-2} = 0.6667$$

$$E(\theta|D) = \frac{a+H}{a+b+n} = \frac{2+7}{2+2+10} = 0.6429$$

(b)

$$\hat{\phi}_{ML} = \frac{70}{100} = 0.70$$

$$\phi_{MAP} = \frac{2+70-1}{2+2+100-2} = \frac{71}{102} = 0.6961$$

Posterior mean: $= \frac{2+70}{2+2+100} = \frac{72}{104} = 0.6923$

The Beta(2,2) prior pulls estimates towards 0.5. Its influence decreases as the number of observations increases.

12) Hospital Sepsis Early-Warning System

(a)

Component	Description
Task (T)	Predict sepsis without the next 6 hours
Performance (P)	Sensitivity, specificity, F1-score
Experience (E)	Historical electronic health records
Target function	Map patient data to sepsis risk
Input representation	Vital signs, lab values and patient records
Learning Algorithm	Supervised classification using Random Fores
Validation	Stratified train-test split and cross-validation.

(b) $TP = 0.90(50) = 45$
$FN = 50 - 45 = 5$
} Sensitivity = 0.90

$TN = 0.80(950) = 760$
$FP = 950 - 760 = 190$
} Sensitivity = 0.80

Metric	Value
TP	45
FN	5
FP	190
TN	760

(c) $$PPV = \frac{TP}{TP+FP} = \frac{45}{45+190} = 19.15\%$$

$$NPV = \frac{FN}{TN+FN} = \frac{760}{760+5} = 99.35\%$$

(d) Using the model : Cost = $(FN \times 100) + (FP \times 5) = (5 \times 100) + (190 \times 5)$
= ₹1,450

Alerting every patient : Cost = 950×5 = ₹4,750

Never alerting : Cost = 50×100 = ₹5,000

(e) Deployment policy.

- Deploy the model with continuous patient monitoring.
- Set the alert threshold using Bayes decision theory.
- Alert when :

$$100p > 5(1-p)$$
$$150p > 5$$
$$p > 0.0476 = 4.76\%$$

- Use a low threshold to minimise missed sepsis cases
- Validate alerts clinically before deployment.